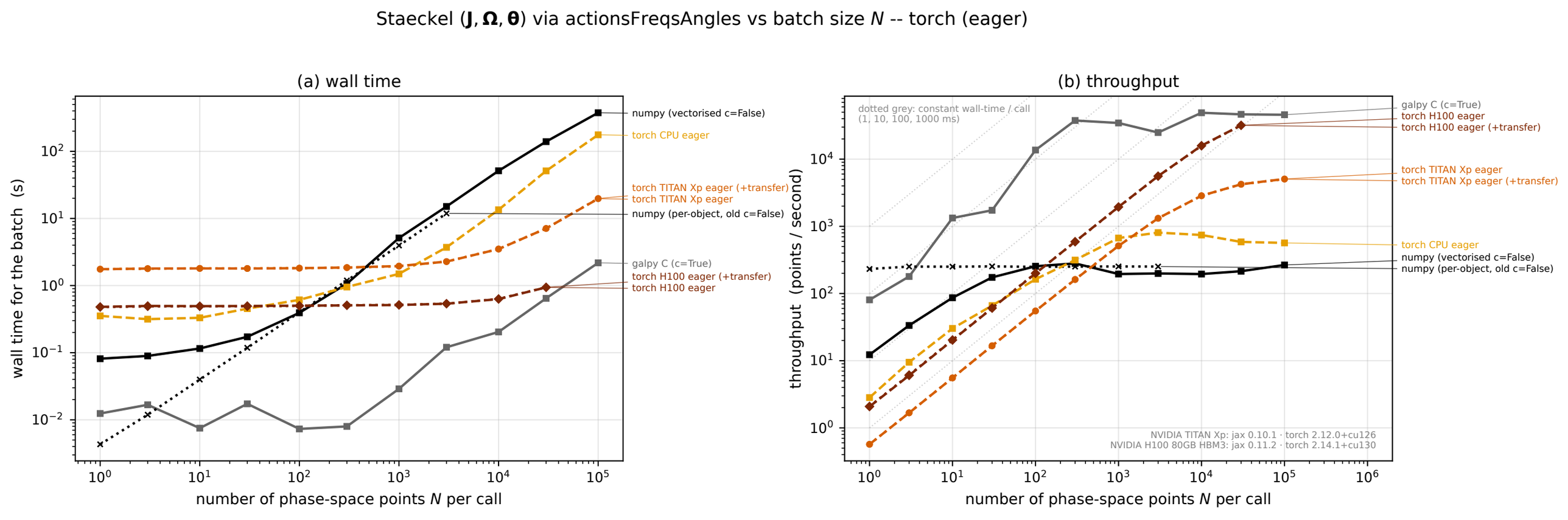
Staeckel ($\mathbf{J}, \mathbf{\Omega}, \mathbf{\theta}$) via actionsFreqsAngles vs batch size N -- torch (eager)
(a) wall time
wall time for the batch (s)
number of phase-space points N per call
numpy (vectorised c=False)
torch CPU eager
torch TITAN Xp eager (+transfer)
torch TITAN Xp eager
numpy (per-object, old c=False)
galpy C (c=True)
torch H100 eager (+transfer)
torch H100 eager
(b) throughput
throughput (points / second)
number of phase-space points N per call
dotted grey: constant wall-time / call
(1, 10, 100, 1000 ms)
NVIDIA TITAN Xp: jax 0.10.1 · torch 2.12.0+cu126
NVIDIA H100 80GB HBM3: jax 0.11.2 · torch 2.14.1+cu130
galpy C (c=True)
torch H100 eager
torch H100 eager (+transfer)
torch TITAN Xp eager
torch TITAN Xp eager (+transfer)
torch CPU eager
numpy (vectorised c=False)
numpy (per-object, old c=False)